01 MCU / RTC / UI

U1
STM32U535CEU6
VDD1 1
VBAT 2
PC13 3
PC14 4
PC15 5
PH0 6
PH1 7
NRST 8
VDDA 9
VSSA_R 10
BAT_ADC 11
MIC_WAKE 12
MIC_PASS 13
SD_CS_N 14
SD_CLK_MCU 15
SD_MOSI 16
SD_MISO 17
SD_EN 18
PB3 19
CHG_DISABLE 20
VCAP 21
GND 22
VDD3 23
MIC_LED 24
PB13 25
PB14 26
PB15 27
PA8 28
PA9 29
PA10 30
PA11 31
PA12 32
PA13 33
VCC 34
GND 35
PA14 36
PA15 37
PB4 38
PB5 39
PB6 40
PB7 41
BOOT0 42
PB8 43
PB9 44
GND 45
VDD 46
BUTTON_N
CHG_N
STATUS_LED
USB_DM_MCU
USB_DP_MCU
SWDIO
GND
VDD3
SWCLK
GND
PDM_CLK_MCU
PDM_DATA
MIC_EN
BOOT0
GND
VDD3
GND

C1
100n
VDD3 1
2 GND

C4
10u
VDD3 1
2 GND

C9
100n
NRST 1
2 GND

R2
10k
VDD3 1
2 NRST

Y1
32.768kHz CL=7pF TBD
LSE_IN 1
2 LSE_OUT

C11
10p tentative
LSE_IN 1
2 GND

C12
10p tentative
LSE_OUT 1
2 GND

J3
TC2030 SWD target pads
VDD3 1 VTref
SWDIO 2 SWDIO
NRST 3 nRESET
4 SWCLK
5 GND
6 SWO_unused

SW1
B3U-1000P (candidate)
BUTTON_N 1
2 GND

R4
100k
VDD3 1
2 BUTTON_N

R5
3.3k
MIC_LED 1
2 LED_A

D1
red LED
LED_A A
K 1 GND

C2
100n
VDD3 1
2 GND

C3
100n
VDD3 1
2 GND

C5
4.7u
VCAP 1
2 GND

C6
100n
VDDA 1
2 GND

C7
1u
VDDA 1
2 GND

R1
0R
VDD3 1
2 VDDA

R3
100k
BOOT0 1
2 GND

02 USB-C / CHARGER

J1
USB4105-GF-A (candidate)
GND A1 GND
VBUS A4 VBUS
CC1 A5 CC1
USB_DP A6 D+
USB_DM A7 D−
SBU1 B8
VBUS B9 VBUS
GND B11 GND
VBUS B9
CC2 B5 CC2
USB_DP B6 D+
SBU2 B8
VBUS B9
GND B12 GND
SHIELD S1 GND

R6
5.1k 1%
CC1 1
2 GND

R7
5.1k 1%
CC2 1
2 GND

R8
0R tuning
USB_DP 1
2 USB_DP_MCU

R9
0R tuning
USB_DM 1
2 USB_DM_MCU

U2
USBLC6-2P6
USB_DP 1 IO1
GND 2 GND
USB_DM 3 IO2
IO2 6 USB_DM
VBUS 5 VBUS
IO1 4 USB_DP

C13
2.2u 10V
VBUS 1
2 GND

U3
BQ25101YFPR
VBUS A2 IN
BAT A1 OUT
ISET B1 ISET
TS B2 BAT_TS
CHG C2 CHG_N
VSS C1 GND

R10
6.19k 1%
ISET 1
2 GND

R11
100k
VDD3 1
2 CHG_N

C14
2.2u
BAT 1
2 GND

J2
078173003 / 0TP401525-XE
BAT 1 PACK+
BAT_TS 2 PACK_NTC
PACK− 7 GND

Q2
2N7002 (charge inhibit)
CHG_DISABLE 1 G
GND 2 S
D 3 BAT_TS

R12
100k
CHG_DISABLE 1
2 GND

Q3
2N7002 (VBUS detect)
VBUS_GATE 1 G
GND 2 S
D 3 VBUS_N

R13
100k
VBUS 1
2 VBUS_GATE

R14
1M
VBUS_GATE 1
2 GND

R15
100k
VDD3 1
2 VBUS_N

#FLG01
PWR_FLAG
GND 1 pwr

#FLG02
PWR_FLAG
VDDA 1 pwr

#FLG03
PWR_FLAG
VBUS 1 pwr

#FLG04
PWR_FLAG
SYS 1 pwr

03 POWER / FUEL GAUGE

D2
B6T54WS (candidate: leakage/forward drop review pending)
VBUS 1
4 SYS

Q1
DMP2035U (candidate: power path review pending)
VBUS 1 S
SYS 2 S
D 3 BAT

R16
100k
VBUS 1
2 GND

C15
4.7u
SYS 1
2 GND

U4
TPS7A0233PDBVR
SYS 1 IN
GND 2 GND
SYS 3 EN
NC 4 NC
OUT 5 VDD3

U5
TPS7A0218PDBVR
VDD3 1 IN
GND 2 GND
MIC_EN 3 EN
NC 4 NC
OUT 5 VDDB

C16
1u
VDD3 1
2 GND

C18
1u
VDDB 1
2 GND

C19
1u
VDD3 1
2 GND

R17
100k
MIC_EN 1
2 GND

TP1
TestPoint
BAT 1 BAT

TP2
TestPoint
SYS 1 SYS

TP3
TestPoint
VDD3 1 3V3

TP4
TestPoint
VDDB 1 VDDB

TP5
TestPoint
SD_V3 1 SD_V3

TP6
TestPoint
GND 1 GND

R33
1M 1%
BAT 1
2 BAT_ADC

R34
1M 1%
BAT_ADC 1
2 GND

C28
10n
BAT_ADC 1
2 GND

04 AUDIO / SD / INDICATORS

U8
TS4230
MIC_DATA_EXR 1 DATA
GND 2 SELECT
GND 3 GND
MIC_WAKE_EXR 4 WAKE
5 MIC_THSEL_EXR
6 CLK
7 VDDB

U9
TXS0104ERUTR
VDDB 1 VCCA
MIC_WAKE_EXR 2 A1
MIC_THSEL_EXR 3 A2
MIC_DATA_EXR 4 A3
MIC_CLK_EXR 5 A4
GND 6 GND
12 VCCB
11 B4
10 B3
9 B2
8 B1
7 VDDB
MIC_CLK_EXR
PDM_DATA
MIC_THSEL
MIC_WAKE

R21
22R tentative
PDM_CLK_MCU 1
2 PDM_CLK

C22
100n
VDDB 1
2 GND

C23
100n
VDDB 1
2 GND

C24
100n
VDD3 1
2 GND

U10
TPS22906CYFFR
VDD3 A2 VIN
SD_V3 A1 VOUT
ON B2
GND B1
SD_EN
GND

R22
100k
SD_EN 1
2 GND

C25
1u
VDD3 1
2 GND

U11
CSNP32GCR01-BOW
3V 1 3V
SD_CS_N 2 CS_N
SD_CLK 3 CLK
VSS 4 VSS
5 DI
DO 6
RSV 7
VCC 8
SD_MOSI
SD_MISO
SD_V3

R23
22R tentative
SD_CLK_MCU 1
2 SD_CLK

R24
47k
SD_V3 1
2 SD_MOSI

R25
47k
SD_V3 1
2 SD_MISO

R28
47k
SD_V3 1
2 SD_CS_N

C26
100n
SD_V3 1
2 GND

C27
22u effective TBD
SD_V3 1
2 GND

R32
6.8k tentative
STATUS_LED 1
2 STATUS_LED_A

D6
amber LED 0402 TBD
STATUS_LED_A A
K 1 GND

Unrouted PCB / no manufacturing release

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Title: Pendant Recorder — Omi-derived electrical design

Size: A3 | Date: 2026-10-24 | Rev: rev-e DRAFT
KiCad 8.0.4 10.0.1 | Id: 1/1